As you read this section, you'll likely make comparisons to Ruby modules and classes. Elixir modules are similar to Ruby modules because they each serve the purpose of organization. However, Elixir modules offer hook points that can be used for metaprogramming. This makes them similar to classes as well. It's best to acknowledge the similarities in name and syntax but accept that Ruby and Elixir have their differences.

Before we write a module, we need an easier way to run our code. Manually typing code into an IEx session is a bit cumbersome. Luckily, Elixir has Mix.

Mix Projects

Mix[2] is a tool that helps us create, manage, compile, and deploy Elixir applications. Plus, it manages our dependencies for us too! It ships with Elixir, so you already have it available. It's similar to Ruby's Rake tool combined with Bundler.

You could compile and execute Elixir files yourself, but it would be a pretty messy process. You would need to worry about dependencies, loading the compiled files, and then executing a script. Mix handles all of this for us. An application that uses Mix is referred to as a Mix project.

The next few chapters use a Mix project to compile longer sections of code. You're going to create that project now, but make sure to put it in a convenient directory because you'll be coming back to it several times.

Run the following command in a directory that you want to create the Mix project in:

```
$ mix new examples
$ cd examples
$ mix test
```

Nice! You just made a new Mix project called `examples`. This will allow you to easily run the scripts in this chapter. If you saw green when you ran `mix test`, then everything is set up properly. If you see red, then check to make sure that your Elixir and Erlang versions are installed correctly.

Inside of the newly created `examples` directory, you'll see `lib` and `test` folders. Application code goes into the `lib` folder and tests go into the `test` folder. Technically, you could implement a different code organization, but it's extremely rare to see any other type of code structure in Elixir applications.

Now we can write some Elixir!

2. https://hexdocs.pm/mix/Mix.html